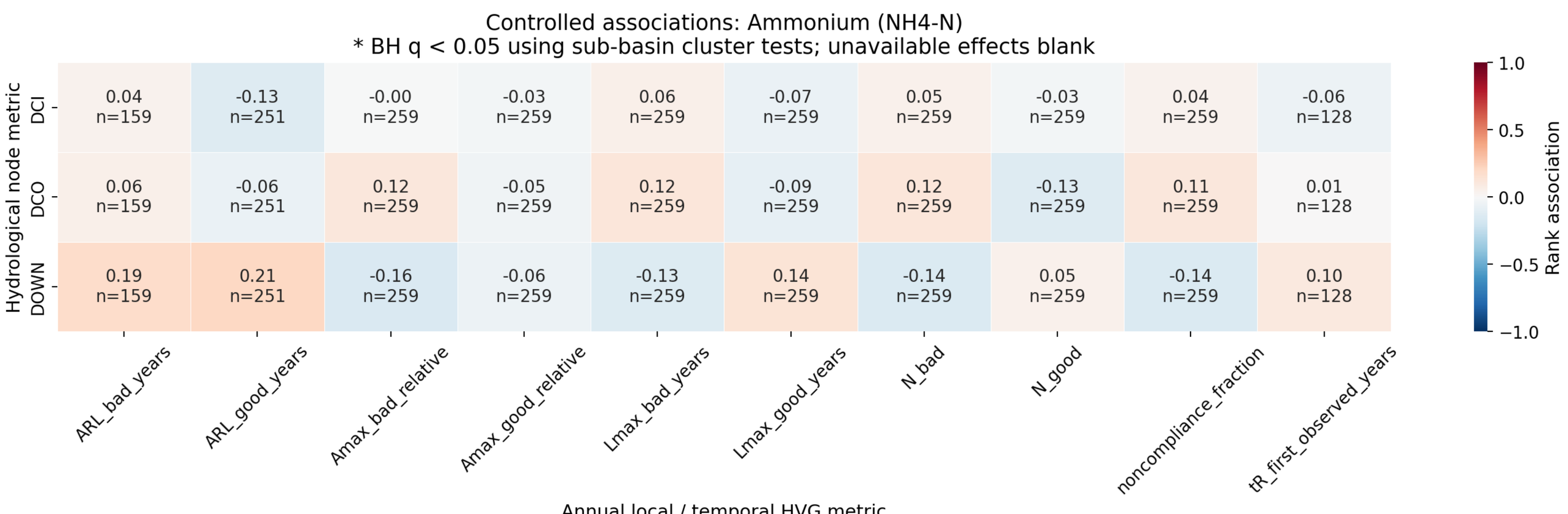
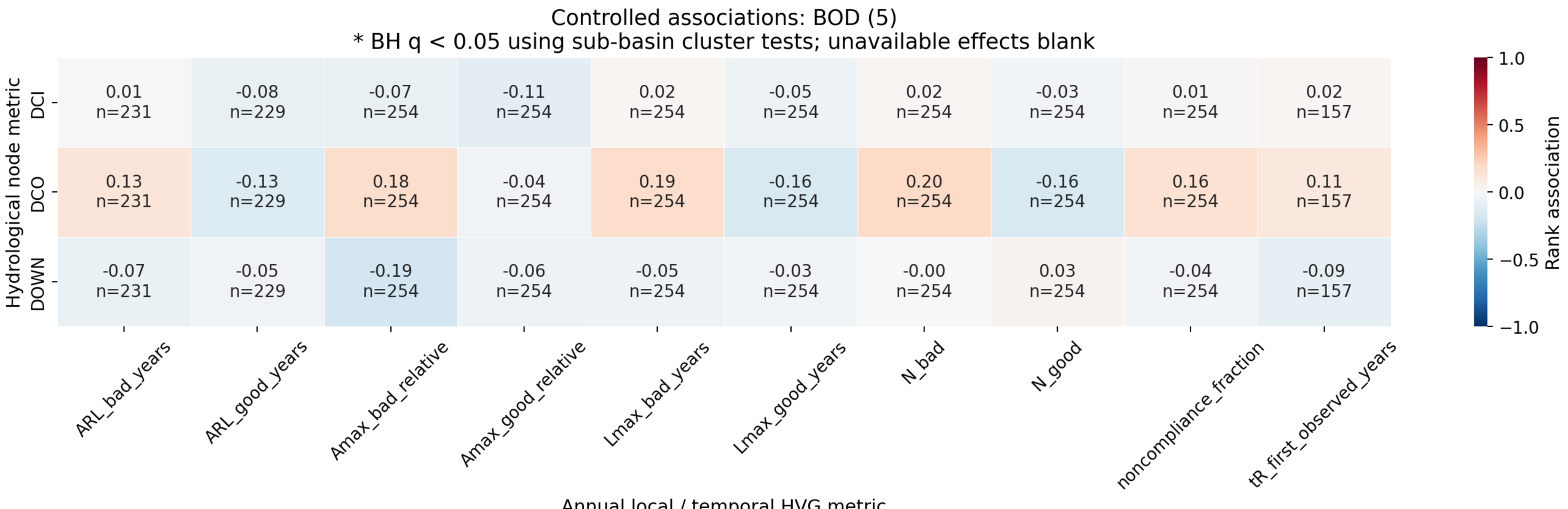
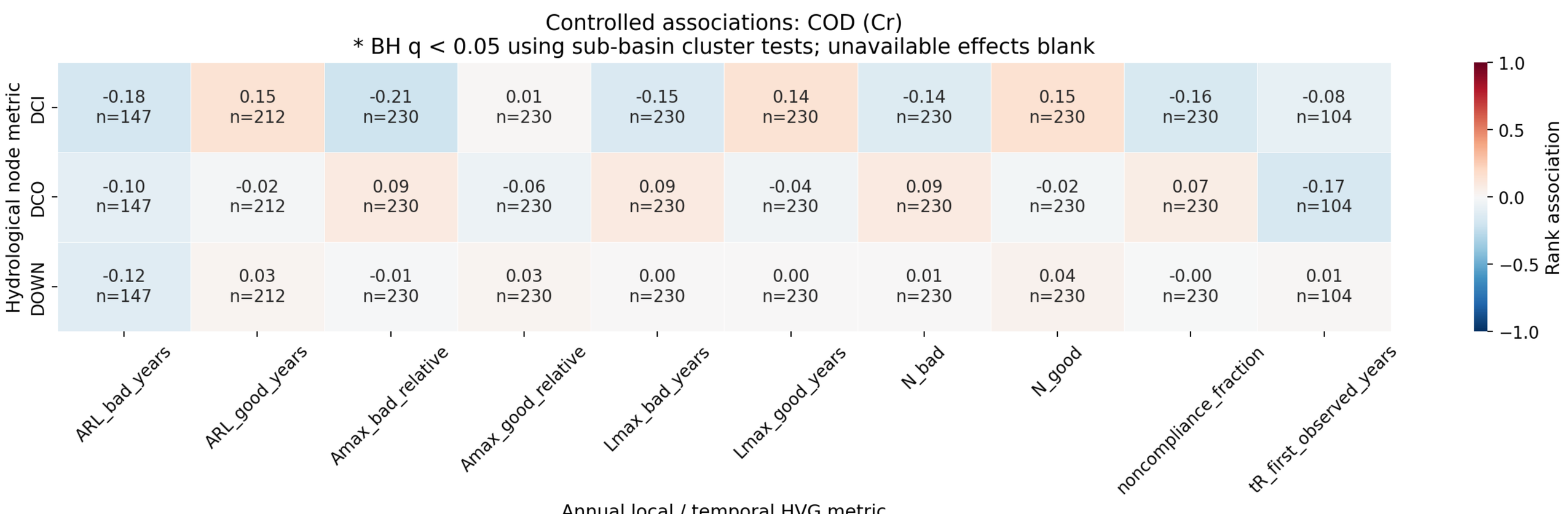
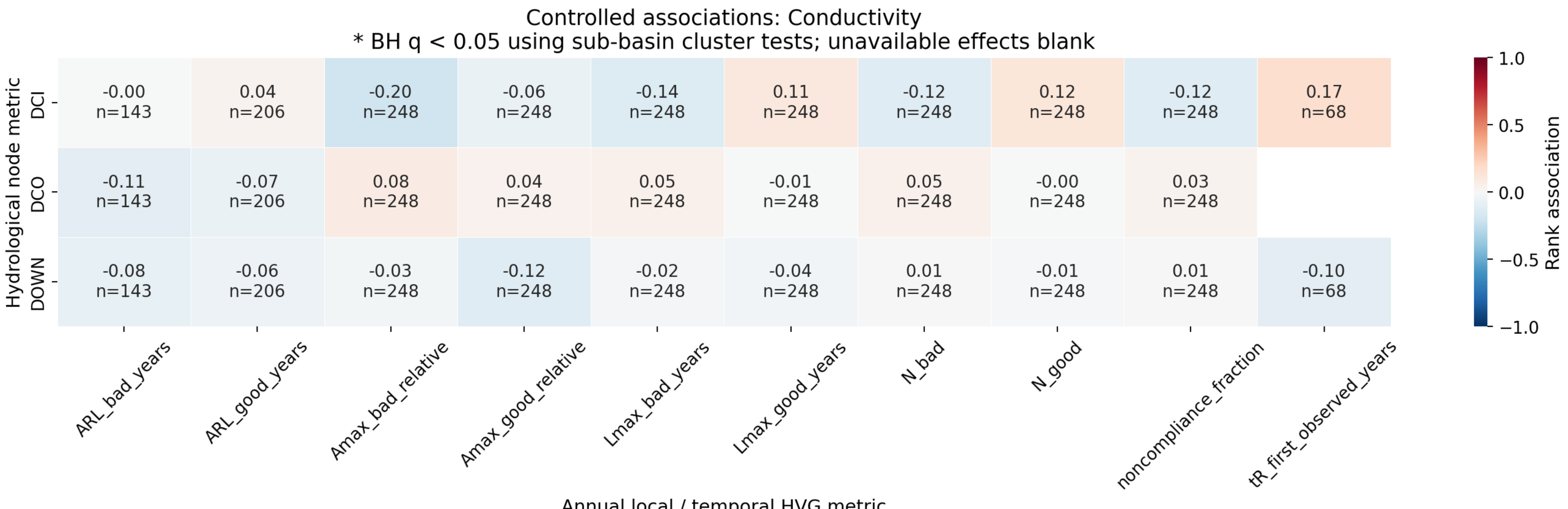
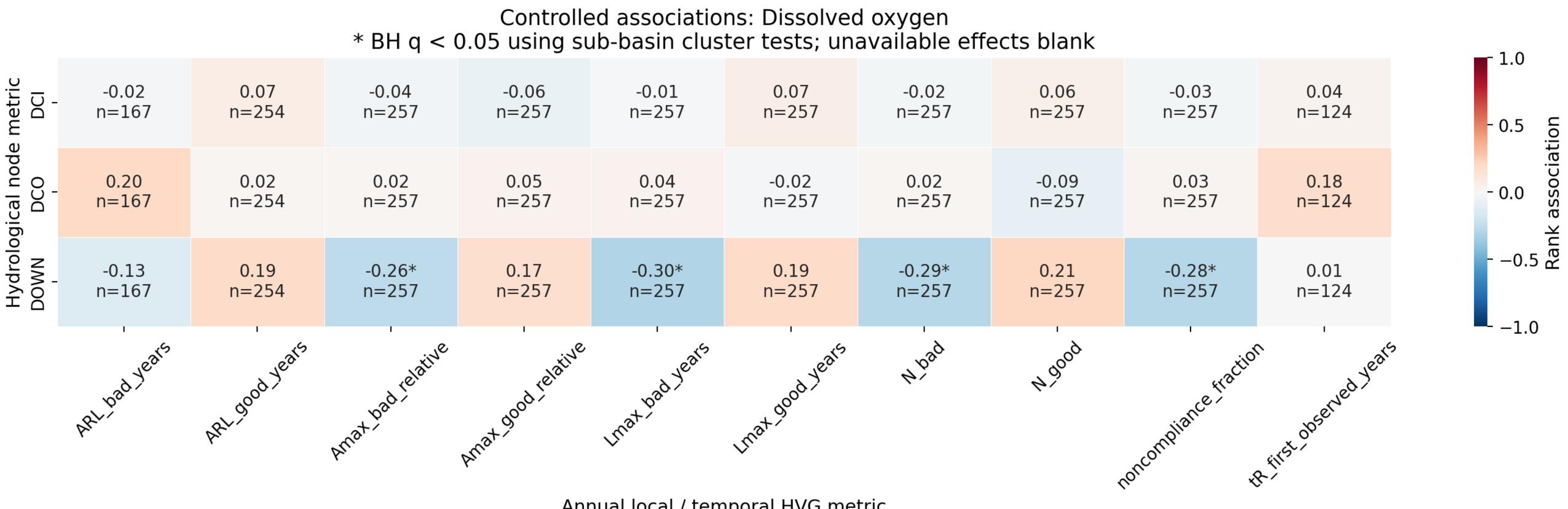
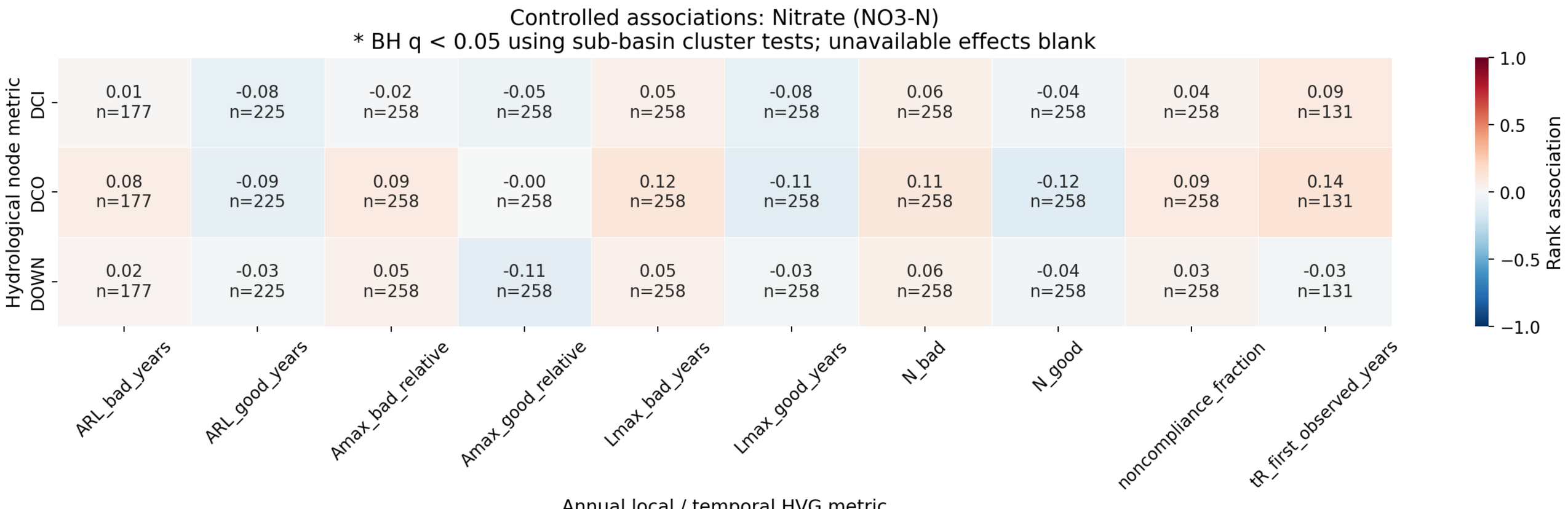

Controlled associations: Ammonium (NH4-N)
* BH q < 0.05 using sub-basin cluster tests; unavailable effects blank
Hydrological node metric
DCI
DCO
DOWN
0.04 n=159
-0.13 n=251
-0.00 n=259
-0.03 n=259
0.06 n=259
-0.07 n=259
0.05 n=259
-0.03 n=259
0.04 n=259
-0.06 n=128
0.06 n=159
-0.06 n=251
0.12 n=259
-0.05 n=259
0.12 n=259
-0.09 n=259
0.12 n=259
-0.13 n=259
0.11 n=259
0.01 n=128
0.19 n=159
0.21 n=251
-0.16 n=259
-0.06 n=259
-0.13 n=259
0.14 n=259
-0.14 n=259
0.05 n=259
-0.14 n=259
0.10 n=128
ARL_bad_years
ARL_good_years
Amax_bad_relative
Amax_good_relative
Lmax_bad_years
Lmax_good_years
N_bad
N_good
noncompliance_fraction
tR_first_observed_years
Annual local / temporal HVG metric
Rank association
1.0
0.5
0.0
−0.5
−1.0

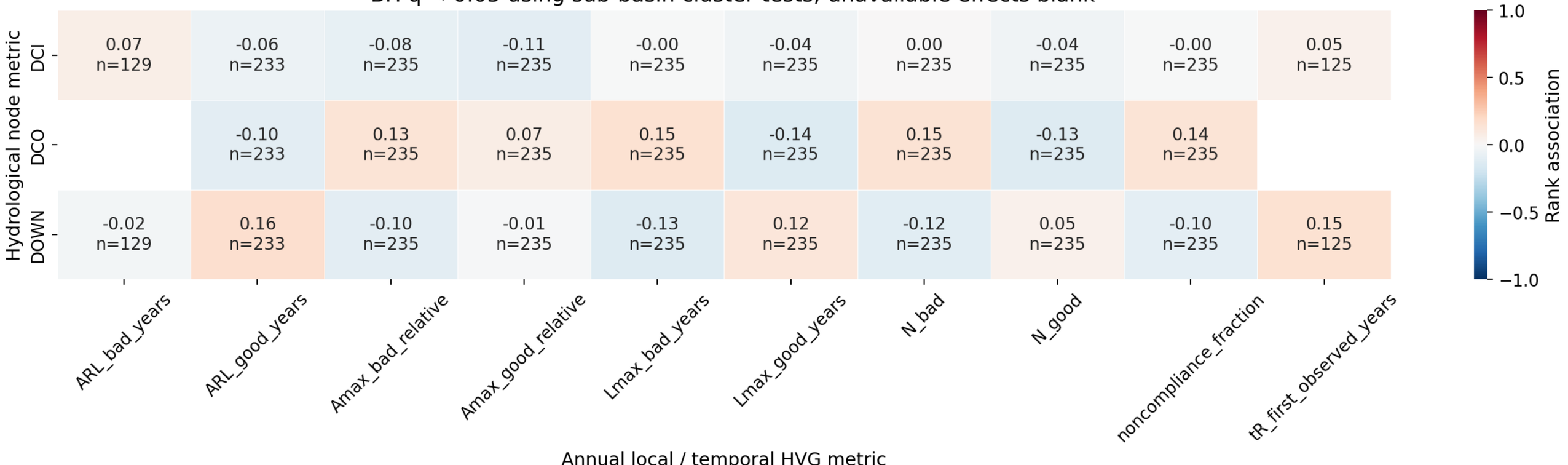

Controlled associations: BOD (5)
* BH q < 0.05 using sub-basin cluster tests; unavailable effects blank
Hydrological node metric
DCI
DCO
DOWN
0.01 n=231
-0.08 n=229
-0.07 n=254
-0.11 n=254
0.02 n=254
-0.05 n=254
0.02 n=254
-0.03 n=254
0.01 n=254
0.02 n=157
0.13 n=231
-0.13 n=229
0.18 n=254
-0.04 n=254
0.19 n=254
-0.16 n=254
0.20 n=254
-0.16 n=254
0.16 n=254
0.11 n=157
-0.07 n=231
-0.05 n=229
-0.19 n=254
-0.06 n=254
-0.05 n=254
-0.03 n=254
-0.00 n=254
0.03 n=254
-0.04 n=254
-0.09 n=157
ARL_bad_years
ARL_good_years
Amax_bad_relative
Amax_good_relative
Lmax_bad_years
Lmax_good_years
N_bad
N_good
noncompliance_fraction
tR_first_observed_years
Annual local / temporal HVG metric
Rank association
1.0
0.5
0.0
−0.5
−1.0

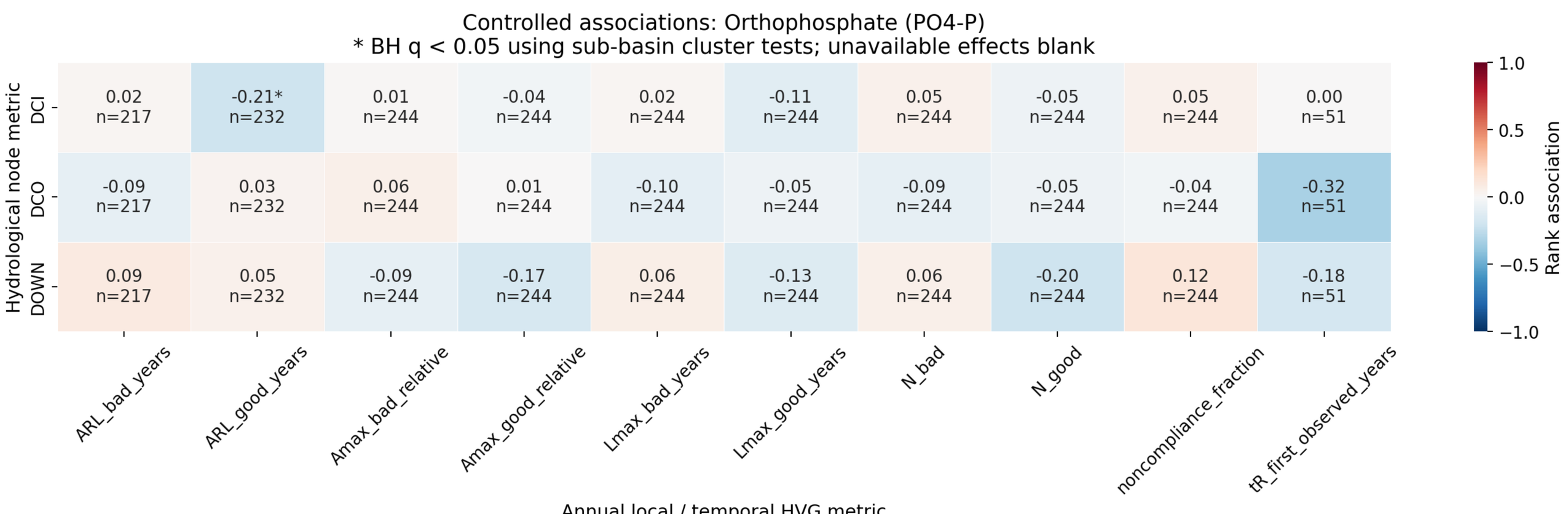

Controlled associations: COD (Cr)
* BH q < 0.05 using sub-basin cluster tests; unavailable effects blank
Hydrological node metric
DCI
DCO
DOWN
-0.18 n=147
0.15 n=212
-0.21 n=230
0.01 n=230
-0.15 n=230
0.14 n=230
-0.14 n=230
0.15 n=230
-0.16 n=230
-0.08 n=104
-0.10 n=147
-0.02 n=212
0.09 n=230
-0.06 n=230
0.09 n=230
-0.04 n=230
0.09 n=230
-0.02 n=230
0.07 n=230
-0.17 n=104
-0.12 n=147
0.03 n=212
-0.01 n=230
0.03 n=230
0.00 n=230
0.00 n=230
0.01 n=230
0.04 n=230
-0.00 n=230
0.01 n=104
ARL_bad_years
ARL_good_years
Amax_bad_relative
Amax_good_relative
Lmax_bad_years
Lmax_good_years
N_bad
N_good
noncompliance_fraction
tR_first_observed_years
Annual local / temporal HVG metric
Rank association
1.0
0.5
0.0
−0.5
−1.0

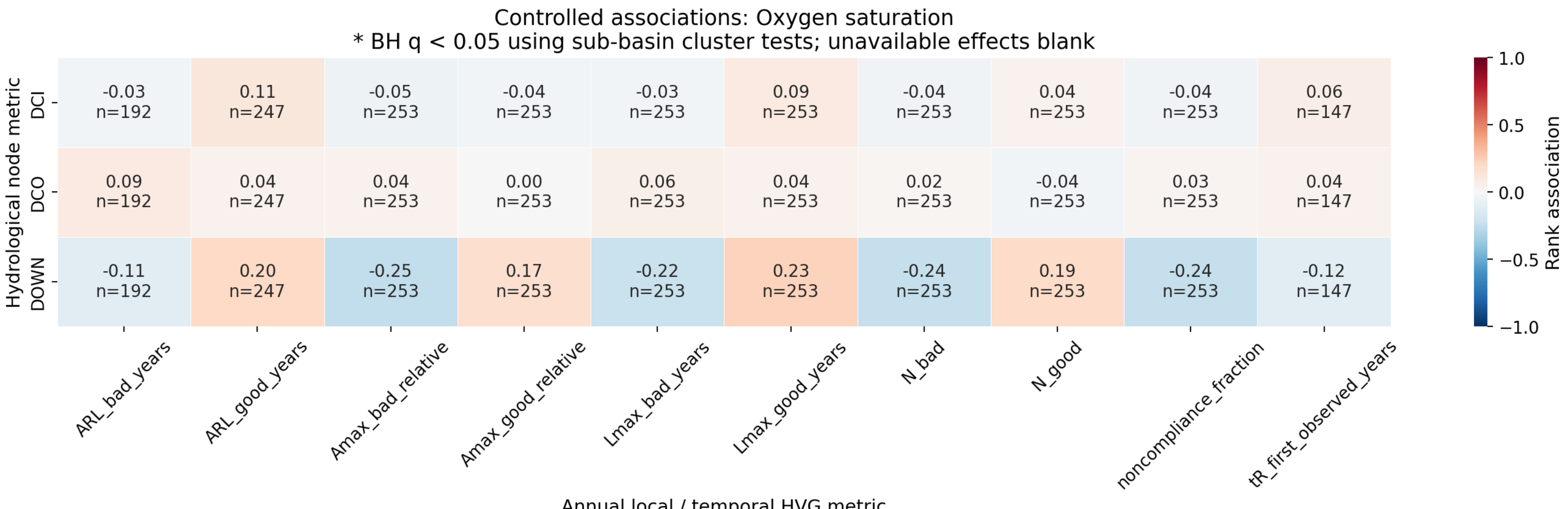

Controlled associations: Conductivity
* BH q < 0.05 using sub-basin cluster tests; unavailable effects blank
Hydrological node metric
DCI
DCO
DOWN
-0.00 n=143
0.04 n=206
-0.20 n=248
-0.06 n=248
-0.14 n=248
0.11 n=248
-0.12 n=248
0.12 n=248
-0.12 n=248
0.17 n=68
-0.11 n=143
-0.07 n=206
0.08 n=248
0.04 n=248
0.05 n=248
-0.01 n=248
0.05 n=248
-0.00 n=248
0.03 n=248
-0.08 n=143
-0.06 n=206
-0.03 n=248
-0.12 n=248
-0.02 n=248
-0.04 n=248
0.01 n=248
-0.01 n=248
0.01 n=248
-0.10 n=68
ARL_bad_years
ARL_good_years
Amax_bad_relative
Amax_good_relative
Lmax_bad_years
Lmax_good_years
N_bad
N_good
noncompliance_fraction
tR_first_observed_years
Annual local / temporal HVG metric
Rank association
1.0
0.5
0.0
−0.5
−1.0

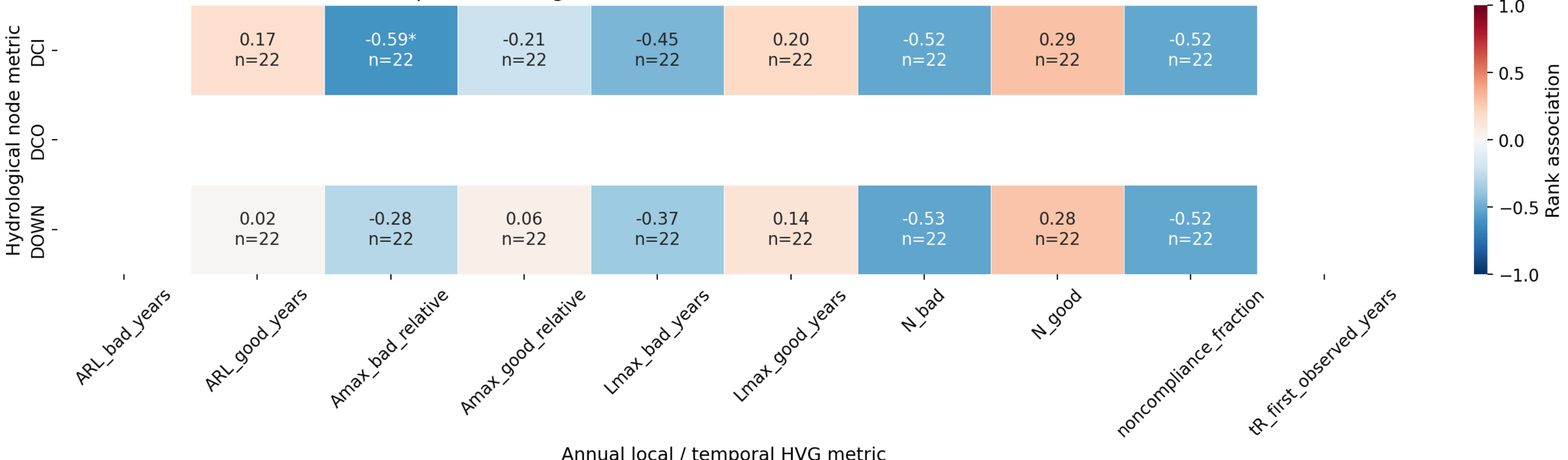

Controlled associations: Dissolved oxygen
* BH q < 0.05 using sub-basin cluster tests; unavailable effects blank
Hydrological node metric
DCI
DCO
DOWN
-0.02 n=167
0.07 n=254
-0.04 n=257
-0.06 n=257
-0.01 n=257
0.07 n=257
-0.02 n=257
0.06 n=257
-0.03 n=257
0.04 n=124
0.20 n=167
0.02 n=254
0.02 n=257
0.05 n=257
0.04 n=257
-0.02 n=257
0.02 n=257
-0.09 n=257
0.03 n=257
0.18 n=124
-0.13 n=167
0.19 n=254
-0.26* n=257
0.17 n=257
-0.30* n=257
0.19 n=257
-0.29* n=257
0.21 n=257
-0.28* n=257
0.01 n=124
ARL_bad_years
ARL_good_years
Amax_bad_relative
Amax_good_relative
Lmax_bad_years
Lmax_good_years
N_bad
N_good
noncompliance_fraction
tR_first_observed_years
Annual local / temporal HVG metric
Rank association
1.0
0.5
0.0
−0.5
−1.0

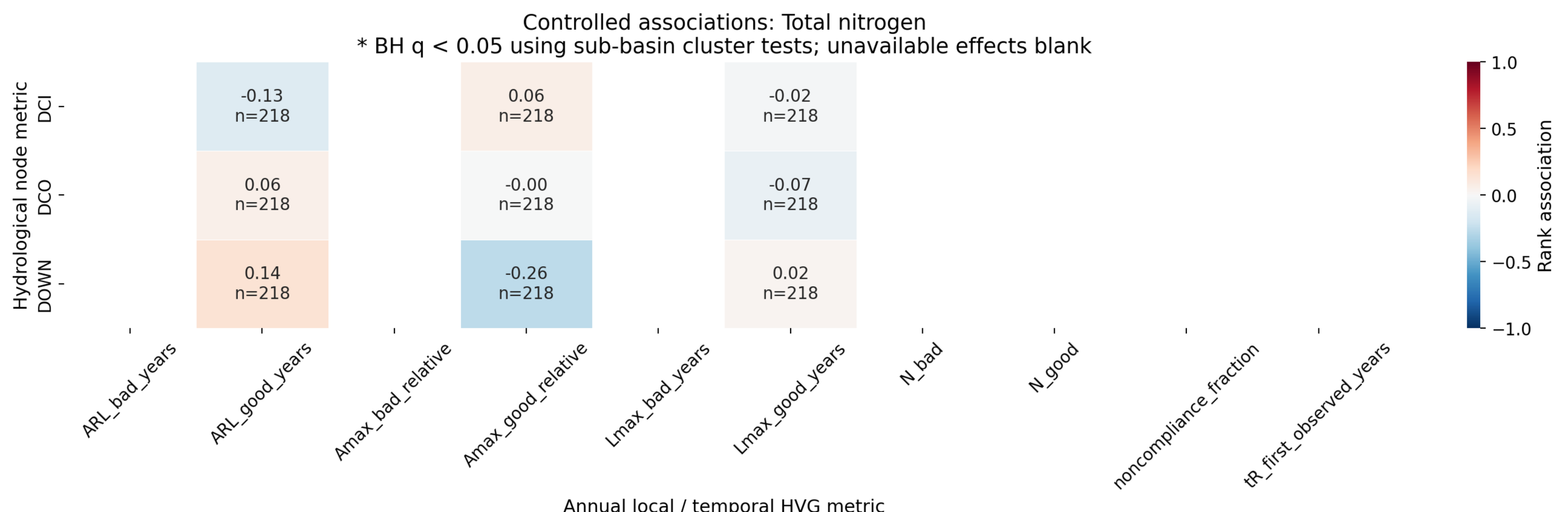

Controlled associations: Nitrate (NO3-N)
* BH q < 0.05 using sub-basin cluster tests; unavailable effects blank
Hydrological node metric
DCI
DCO
DOWN
0.01 n=177
-0.08 n=225
-0.02 n=258
-0.05 n=258
0.05 n=258
-0.08 n=258
0.06 n=258
-0.04 n=258
0.04 n=258
0.09 n=131
0.08 n=177
-0.09 n=225
0.09 n=258
-0.00 n=258
0.12 n=258
-0.11 n=258
0.11 n=258
-0.12 n=258
0.09 n=258
0.14 n=131
0.02 n=177
-0.03 n=225
0.05 n=258
-0.11 n=258
0.05 n=258
-0.03 n=258
0.06 n=258
-0.04 n=258
0.03 n=258
-0.03 n=131
ARL_bad_years
ARL_good_years
Amax_bad_relative
Amax_good_relative
Lmax_bad_years
Lmax_good_years
N_bad
N_good
noncompliance_fraction
tR_first_observed_years
Annual local / temporal HVG metric
Rank association
1.0
0.5
0.0
−0.5
−1.0

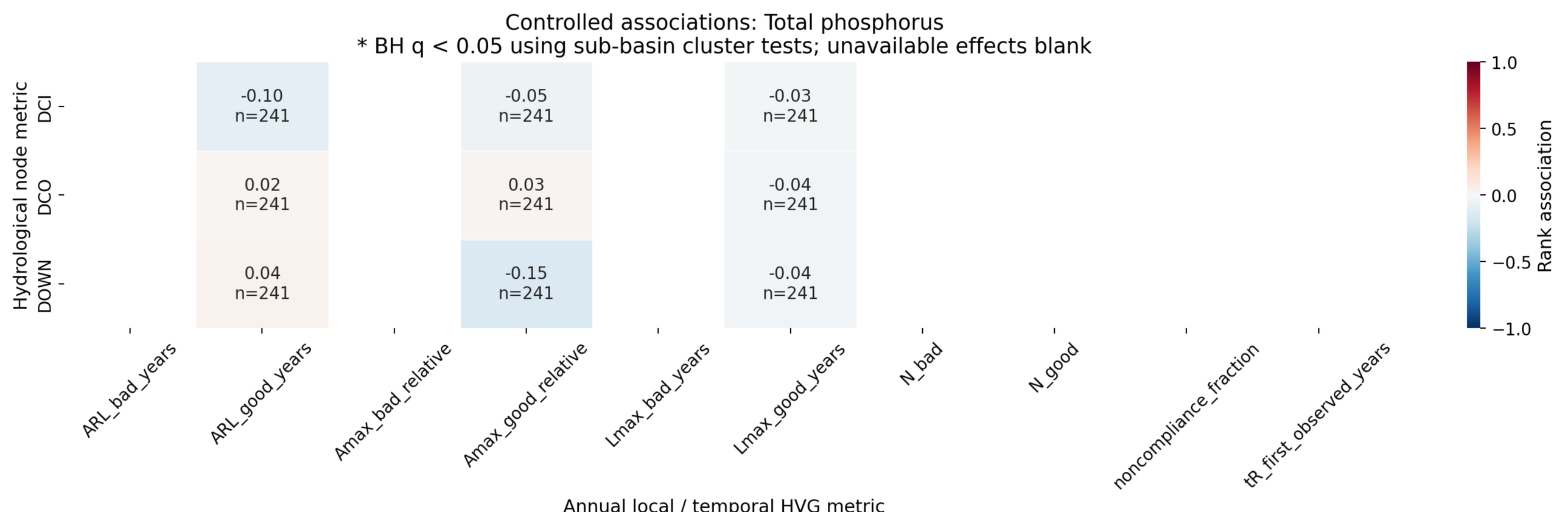

Controlled associations: Nitrite (NO2-N)
* BH q < 0.05 using sub-basin cluster tests; unavailable effects blank
Hydrological node metric
DCI
DCO
DOWN
0.07 n=129
-0.06 n=233
-0.08 n=235
-0.11 n=235
-0.00 n=235
-0.04 n=235
0.00 n=235
-0.04 n=235
-0.00 n=235
0.05 n=125
-0.10 n=233
0.13 n=235
0.07 n=235
0.15 n=235
-0.14 n=235
0.15 n=235
-0.13 n=235
0.14 n=235
-0.02 n=129
0.16 n=233
-0.10 n=235
-0.01 n=235
-0.13 n=235
0.12 n=235
-0.12 n=235
0.05 n=235
-0.10 n=235
0.15 n=125
ARL_bad_years
ARL_good_years
Amax_bad_relative
Amax_good_relative
Lmax_bad_years
Lmax_good_years
N_bad
N_good
noncompliance_fraction
tR_first_observed_years
Annual local / temporal HVG metric
Rank association
1.0
0.5
0.0
−0.5
−1.0


Controlled associations: Orthophosphate (PO4-P)
* BH q < 0.05 using sub-basin cluster tests; unavailable effects blank
Hydrological node metric
DCI
DCO
DOWN
0.02
n=217
-0.21*
n=232
0.01
n=244
-0.04
n=244
0.02
n=244
-0.11
n=244
0.05
n=244
-0.05
n=244
0.05
n=244
0.00
n=51
-0.09
n=217
0.03
n=232
0.06
n=244
0.01
n=244
-0.10
n=244
-0.05
n=244
-0.09
n=244
-0.05
n=244
-0.04
n=244
-0.32
n=51
0.09
n=217
0.05
n=232
-0.09
n=244
-0.17
n=244
0.06
n=244
-0.13
n=244
0.06
n=244
-0.20
n=244
0.12
n=244
-0.18
n=51
ARL_bad_years
ARL_good_years
Amax_bad_relative
Amax_good_relative
Lmax_bad_years
Lmax_good_years
N_bad
N_good
noncompliance_fraction
tR_first_observed_years
Annual local / temporal HVG metric
Rank association
1.0
0.5
0.0
−0.5
−1.0


Controlled associations: Oxygen saturation
* BH q < 0.05 using sub-basin cluster tests; unavailable effects blank
Hydrological node metric
DCI
DCO
DOWN
-0.03 n=192
0.11 n=247
-0.05 n=253
-0.04 n=253
-0.03 n=253
0.09 n=253
-0.04 n=253
0.04 n=253
-0.04 n=253
0.06 n=147
0.09 n=192
0.04 n=247
0.04 n=253
0.00 n=253
0.06 n=253
0.04 n=253
0.02 n=253
-0.04 n=253
0.03 n=253
0.04 n=147
-0.11 n=192
0.20 n=247
-0.25 n=253
0.17 n=253
-0.22 n=253
0.23 n=253
-0.24 n=253
0.19 n=253
-0.24 n=253
-0.12 n=147
ARL_bad_years
ARL_good_years
Amax_bad_relative
Amax_good_relative
Lmax_bad_years
Lmax_good_years
N_bad
N_good
noncompliance_fraction
tR_first_observed_years
Annual local / temporal HVG metric
Rank association
1.0
0.5
0.0
−0.5
−1.0


Controlled associations: Phytoplankton (biomass - chlorophyll-a)
* BH q < 0.05 using sub-basin cluster tests; unavailable effects blank
Hydrological node metric
DCI
DCO
DOWN
0.17 n=22
-0.59* n=22
-0.21 n=22
-0.45 n=22
0.20 n=22
-0.52 n=22
0.29 n=22
-0.52 n=22
0.02 n=22
-0.28 n=22
0.06 n=22
-0.37 n=22
0.14 n=22
-0.53 n=22
0.28 n=22
-0.52 n=22
ARL_bad_years
ARL_good_years
Amax_bad_relative
Amax_good_relative
Lmax_bad_years
Lmax_good_years
N_bad
N_good
noncompliance_fraction
tR_first_observed_years
Annual local / temporal HVG metric
Rank association
1.0
0.5
0.0
−0.5
−1.0


Controlled associations: Total nitrogen
* BH q < 0.05 using sub-basin cluster tests; unavailable effects blank
Hydrological node metric
DCI
DCO
DOWN
-0.13
n=218
0.06
n=218
-0.02
n=218
0.06
n=218
-0.00
n=218
-0.07
n=218
0.14
n=218
-0.26
n=218
0.02
n=218
ARL_bad_years
ARL_good_years
Amax_bad_relative
Amax_good_relative
Lmax_bad_years
Lmax_good_years
N_bad
N_good
noncompliance_fraction
tR_first_observed_years
Annual local / temporal HVG metric
Rank association
1.0
0.5
0.0
−0.5
−1.0


Controlled associations: Total phosphorus
* BH q < 0.05 using sub-basin cluster tests; unavailable effects blank
Hydrological node metric
DCI
DCO
DOWN
-0.10
n=241
-0.05
n=241
-0.03
n=241
0.02
n=241
0.03
n=241
-0.04
n=241
0.04
n=241
-0.15
n=241
-0.04
n=241
ARL_bad_years
ARL_good_years
Amax_bad_relative
Amax_good_relative
Lmax_bad_years
Lmax_good_years
N_bad
N_good
noncompliance_fraction
tR_first_observed_years
Annual local / temporal HVG metric
Rank association
1.0
0.5
0.0
−0.5
−1.0